Risk Factors for Hospital Readmission in Patients with Type 2 Diabetes Using Survival Analysis Methods

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Abstract

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The increase in hospital readmissions of patients with type 2 diabetes has incurred substantial costs on health care. Risk factors for readmission include both demographic and clinical characteristics of patients. Recent studies on hospital readmission focus on the 30-day readmission rate, as it is the metric used by the Hospital Readmission Reduction Program to evaluate hospital care quality. However, previous studies have shown that patient risk factors may vary at different time points after discharge from hospital. Understanding the changes in risk factors helps hospitals develop interventions to reduce the readmission rate effectively. We obtained data from the Medical Information Mart for Intensive Care (MIMIC-IV), a clinical database containing de-identified admission records for patients admitted to the Beth Israel Deaconess Medical Center in Boston, Massachusetts. We compared the risk factors for readmission in 30 days, 30-90 days, and 90-180 days. We modeled the time-to-readmission using survival analysis methods such as the proportional hazards and the accelerated failure time models.

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Chapter 1
INTRODUCTION

1.1 Background

As reported by the Centers for Disease Control and Prevention, 34.2 million US population had diabetes in 2018. Type 2 diabetes constitutes about 90 to 95 percent of all diabetes cases. Prevalence of diabetes differed significantly by socioeconomic status, including income and education level. Among all hospitalized diabetes patients, 14.4 to 22.7 percent experience readmissions (Ostling et al., 2017). Hospital readmissions are costly but sometimes preventable. The Centers for Medicare and Medicaid Services launched the Hospital Readmission Reduction Program in 2012, aiming to improve the quality of care in hospitals and reduce avoidable readmissions. Readmission reduction programs and interventions can reduce the total cost of diabetes, which was approximately $327 million in the US in 2017. The design of effective interventions requires an understanding of the risk factors for readmission.

1.2 Literature Review

Corrigan and Martin (1992) categorized risk factors associated with early hospital readmission into patient and care provider characteristics. Patient characteristics further consist of demographic and clinical characteristics. Studies have found that demographic covariates including age, gender, ethnicity, insurance type, and index hospital admission type were associated with readmission events. However, results are inconsistent. Robbins and Webb (2006) reported higher readmission risk in male patients, but Bennett et al. (2012) found that female Medicare beneficiaries were more likely to be readmitted. Results on ethnicity are also conflicting. Although multiple studies have shown an increased risk of 30-day readmission

for Black/African Americans, Jiang et al. (2005) found no difference in early readmission between Black/African Americans and Caucasians. However, Jiang et al. (2005) found an increased likelihood of readmission among Black/African Americans in more extended follow-up periods, suggesting that the difference might attribute to outpatient management rather than inpatient care.

Clinical covariates associated with increased readmission risk include the length of index hospitalization and other disease-related variables. For type 2 diabetes, research has revealed that clinical covariates such as micro and macro vascular complications, comorbidity burden, and pre-admission insulin were correlated with a higher likelihood of readmission events. Eby et al. (2015) found that diabetic treatment escalation strongly correlates to early readmission by comparing prescriptions during index hospitalizations among patients. Discharge location also plays a significant role. According to Wei et al. (2013), patients discharged home with nursing services have a higher readmission risk, since being discharged with additional care may be an indication of higher severity of the disease.

Interventions that effectively reduce readmission are primarily concerned with discharge planning which involves preparations for patients' future health care needs after discharging. Naylor et al. (1999) conducted a randomized clinical trial and found comprehensive discharge planning reduced readmissions and increased the time to readmission in hospitalized elders. Readmission is closely linked to hospital spendings. Rogers et al. (2017) found an association between increased spending in Occupation Therapy and readmission reduction. Inpatient diabetes education can also reduce the frequency of readmission (Healy et al., 2013).

1.3 Research Goals

Recent studies on hospital readmission focus on the 30-day readmission rate as it is the metric used by the Hospital Readmission Reduction Program to evaluate hospital care quality. However, as Jiang et al. (2005) showed, the correlation strength of risk factors may vary at different time points after discharge. Few papers examined the temporal patterns of risk factors. Therefore, one goal of this thesis is to compare readmission risk factors at various

time periods after discharge. An understanding of the changes in risk factors with respect to time is helpful for hospitals to target resources to certain groups of patients at different post-discharge time points. In most existing literature, multiple logistic regression is the primary method. The coefficients from logistic regression can be translated to adjusted odds ratios which have probability representations and are easy to interpret. However, logistic regression requires a binary outcome and can only be used when the time period being considered is fixed. Since our outcome of interest is time-to-readmission, survival analysis methods can be applied to study the underlying hazards and hazard ratios between groups of patients. In chapter 2, we discuss the formulations of the Cox Proportional Hazards (PH) model. We will also discuss the Accelerated Failure Time (AFT) model, which is often considered as an alternative way of modeling time-to-event outcome when the proportional hazards assumption fails, and its applications in the hospital readmission context.

Chapter 2

METHODOLOGY

2.1 Data Source

Data were obtained from the MIMIC-IV (Medical Information Mart for Intensive Care) Clinical Database, a relational database that contains electronic hospital records for patients admitted to the Beth Israel Deaconess Medical Center in Boston, MA, between 2008 and 2019. Data in MIMIC-IV were acquired from two hospital database systems: a hospital-wide EHR and an ICU-specific clinical information system. Access to the database was granted by PhysioNet, which was established under the National Center for Research Resources (NCRR) and the National Institutes of Health (NIH).

MIMIC-IV is an updated version of the MIMIC-III database with more recent hospital admission records and patient data. Researchers have widely used the MIMIC-III database for multiple purposes. Liang et al. (2018) used electrocardiogram (ECG) and photoplethysmogram (PPG) signal data from the MIMIC-III database to predict hypertension. Sandfort et al. (2019) evaluated the correlation between prolonged elevated heart rate and survival in acutely ill patients using heart rate measurements from the same database. Zhao et al. (2020) used platelet level data from the same database as a prognostic predictor for sepsis.

There are currently 524,520 hospital admission records for 383,220 patients in the MIMIC-IV database. Each hospital admission record was assigned a unique hospital admission identification number (HADM ID). All patient data were de-identified, and each patient was assigned a unique identification number (SUBJECT ID). These two identification numbers were used to extract hospital admission data from the database.

2.2 Study Population

Patients admitted to the hospital and who received a type 2 diabetes diagnosis were included. The database recorded medical diagnoses following the International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM). The ICD-10-CM code for type 2 diabetes mellitus begins with "E11." The selected 12,251 patients were divided into two groups, those who had never been readmitted after first discharge (later referred to as the "readmission-free group," $n = 6736$) and those who were readmitted at least once (later referred to as the "readmission group," $n = 6349$).

We obtained the index hospitalization data, including sex, age, insurance type, and ethnicity for patients in the readmission-free group. In addition, a set of clinical covariates including length of stay, admission type, first glucose, and Hemoglobin A1C measures during the index hospitalization were acquired. Note that in admission types, we use "EW" and "EU" to denote Emergency Ward and Emergency Unit respectively. Because studies have suggested that hypertension (HTN), hyperlipidemia (HLD), and chronic kidney disease (CKD) are common comorbidities of type 2 diabetes, we use indicator variables to denote each of those three comorbidities. For patients in the readmission group, in addition to sex, age, insurance type, ethnicity, glucose, A1C, and comorbidities, we obtained the time to first readmission and the same set of clinical covariates at readmission.

2.3 Data Pre-processing

Since the time period of interest starts at the discharge from index hospitalization, patients who died during the index hospitalization were excluded. We also excluded three readmitted cases in which patients' deaths were reported during index hospitalizations. A small number of cases had the same index admission time and readmission time, and they were also removed. The resulting data consisted of 6349 readmission patients and 6736 readmission-free patients. There were 510 cases with missing values for glucose, A1C, or both. With the assumption that missing data were due to absence of lab tests, we performed a chi-squared test

comparing the number of readmission events between patients tested and those not tested. The result indicated that patients tested for glucose during index hospitalization had a higher percentage of readmission events than patients not tested for glucose ($p = 0.034$). However, failure to input test results into the database can also lead to missing values. Therefore, whether data were missing at random is unclear. In our univariate analyses, we included observations with missing data unless the covariate being analyzed was missing. Because the directions of association between covariates and the outcome remained the same and the changes in value of estimates were minor before and after removing observations with missing data, we excluded observations with missing values in the multivariate modeling.

2.4 Hypothesis Tests

We performed a series of hypothesis tests including t-test, Pearson's chi-squared test, Wilcoxon signed-rank test, and log-rank test in section 2.5. In this section, we discuss the Wilcoxon signed-rank test and the log-rank test.

2.4.1 Wilcoxon Signed-Rank Test

Wilcoxon signed-rank test is used to test whether the population mean rank differs between two paired samples. Suppose sample size is n, and let X_i and Y_i where $i = 1, 2, \ldots, n$ be the two paired samples. Then, the test statistic of a Wilcoxon signed-rank test is as follows,

$$W = \sum_{i=1}^{n} \text{sign}(X_i - Y_i) \cdot R_i,$$

where R_i is the rank of $|X_i - Y_i|$. Under the null hypothesis, W converges to a normal distribution with zero mean and variance $n(n+1)(2n+1)/6$. Apart from each paired sample being independent and identically distributed, Wilcoxon signed-rank test also assumes that $|X_i - Y_i|$ has a symmetric distribution.

2.4.2 Log-Rank Test

In the context of hospital readmission, we use log-rank test to compare the distribution of time-to-readmission between two groups of patients. For example, to compare the distribution of time-to-readmission between female and male patients, we construct the following table for every readmission time t,

At time t	Female	Male	Total
Readmitted	$d_0(t)$	$d_1(t)$	$d(t)$
Not-readmitted	$s_0(t)$	$s_1(t)$	$s(t)$
At risk	$n_0(t)$	$n_1(t)$	$n(t)$

In this case, the number of female patients readmitted is equivalent to a hypergeometric random variable. We compute the observed, $o(t)$, and expected, $e(t)$, number of readmitted female patients at time t as

$$o(t) = d_0(t)$$
$$e(t) = \frac{d(t)n_0(t)}{n(t)}.$$

The test statistic is

$$T = \left[\frac{\sum_{i=1}^{n}[o(t_i) - e(t_i)]}{\sqrt{\sum_{i=1}^{n} v(t_i)}}\right]^2,$$

where

$$v(t) = \frac{d(t)s(t)n_0(t)n_1(t)}{n(t)^2[n(t)-1]}.$$

Under the null hypothesis, the distribution of T converges to a chi-squred on 1 degree of freedom.

2.5 Exploratory Analysis

2.5.1 Demographic Covariates

The data consisted of 6776 male patients and 5475 female patients. A chi-squared test on sex and readmission status was performed. The result showed that female patients were more

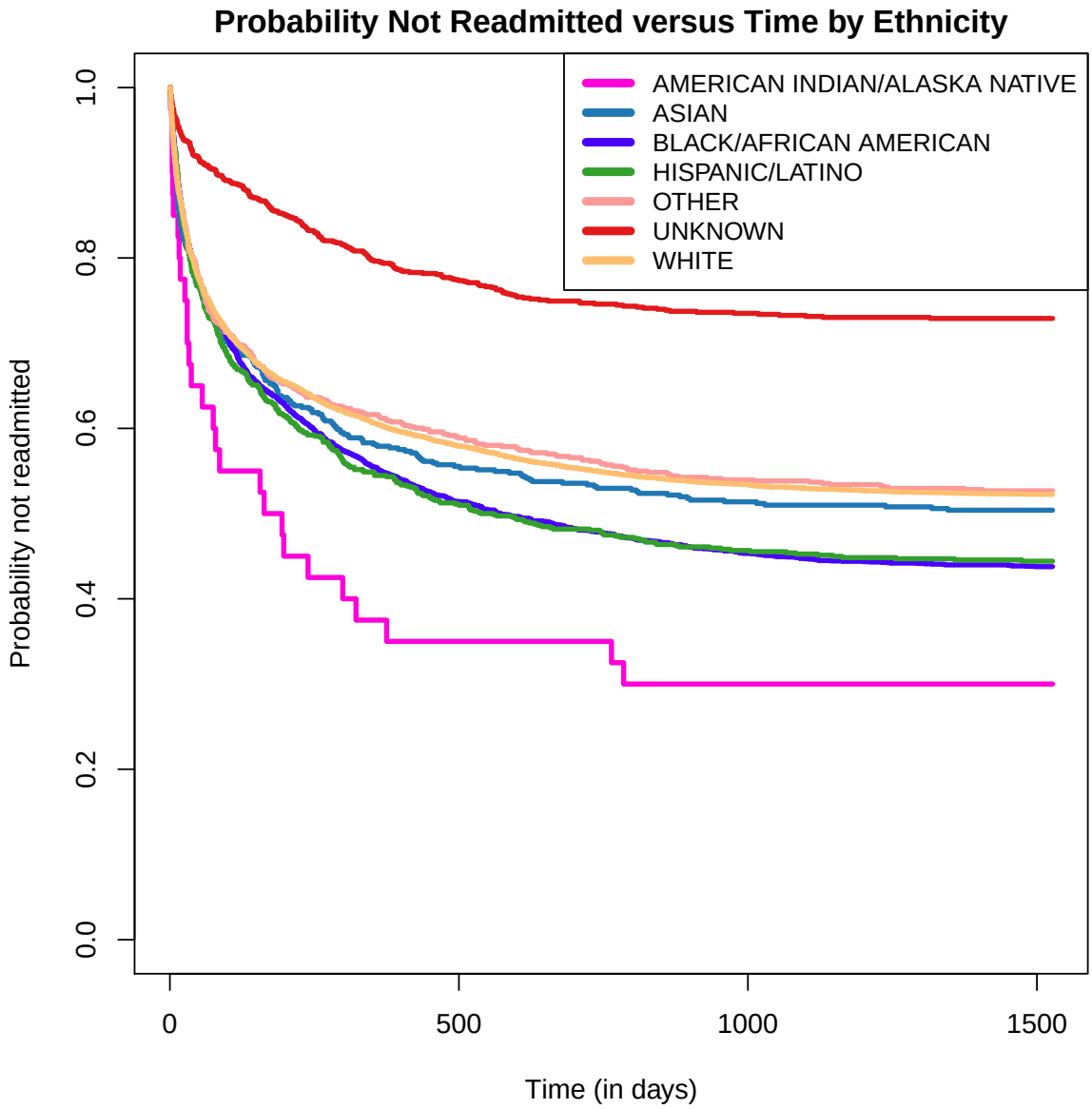
likely to be readmitted than male patients ($p < 0.001$). Both readmission and readmission-free patients had a median age of 69 years old. 4.65% of all patients were enrolled in Medicaid, and 48.2% were enrolled in Medicare at the time of index hospitalization. 16% of Medicare beneficiaries were below 65 years old, as individuals with disabilities or end-stage renal disease in need of kidney transplant are also eligible for Medicare. The median age of Medicaid beneficiaries was 59 years old. Based on a chi-squared test, Medicaid and Medicare beneficiaries had a higher likelihood of readmission than patients with other insurances statuses ($p = 0.003$). Black/African American patients had a higher readmission rate than White patients based on a chi-squared test ($p < 0.001$). The difference in distributions of time-to-readmission between Black and White patients was statistically significant according to a log-rank test ($p < 0.001$). We found no statistically significant difference in distributions of time-to-readmission between Hispanic/Latino and Asian.

To compare the readmission rate across ethnicity groups, we plotted the percentage of patients readmitted versus time after discharge from index hospitalization in Figure 2.1. Notice that the ethnicity group "unknown" had a much lower readmission risk compared to other ethnicity groups. Patients in this group might be lost to follow-up after being discharged from the index hospitalization. The resulted large differences in time-to-readmission can bias inference results. Therefore, we removed observations with "unknown" ethnicity group from our data.

2.5.2 Clinical Covariates

The average length of stay for index admission was longer among readmitted patients than readmission-free patients ($p = 0.007$). This might be because longer length of stay indicates higher severity of the disease. Compared to the readmission-free group, readmitted patients had a lower mean A1C ($p < 0.001$). We found no statistically significant difference in glucose level between the readmission and readmission-free group. Within the readmission group, both the glucose level ($p < 0.001$) and A1C ($p < 0.001$) were lower in readmission than in index hospitalization based on results from Wilcoxon signed-rank tests.

Figure 2.1: Readmission comparison by ethnicity



The median time-to-readmission was 74 days for female and 61 days for male patients. The distributions of time-to-readmission between female and male patients were different according to a log-rank test ($p = 0.006$). Patients with hypertension had a higher mean time-to-readmission than patients without hypertension ($p = 0.038$); patients with hyperlipidemia had a higher mean time-to-readmission than patients without hyperlipidemia ($p = 0.009$); and patients with chronic kidney disease had a higher mean time-to-readmission than patients without chronic kidney disease ($p < 0.001$). Since many patients have multiple comorbidities, we used a Venn diagram (Figure 2.2) to show the number of patients with multiple conditions.

Table 2.1: Summary of categorical covariates

		Readmission Probabilities (95% CI)	
Variable	n	30-day	All-time
Sex			
Female	5810	0.17 (0.16, 0.18)	0.5 (0.49, 0.51)
Male	7275	0.17 (0.16, 0.18)	0.47 (0.46, 0.48)
Insurance			
Medicaid	608	0.21 (0.18, 0.24)	0.5 (0.46, 0.54)
Medicare	6308	0.17 (0.16, 0.18)	0.5 (0.48, 0.51)
Other	6169	0.16 (0.16, 0.17)	0.47 (0.46, 0.48)
Ethnicity			
American Indian/Alaska Native	40	0.3 (0.16, 0.44)	0.7 (0.56, 0.84)
Asian	506	0.19 (0.15, 0.22)	0.5 (0.45, 0.54)
Black/African American	2376	0.18 (0.16, 0.19)	0.56 (0.54, 0.58)
Hispanic/Latino	716	0.19 (0.16, 0.22)	0.56 (0.52, 0.59)
Other	693	0.18 (0.15, 0.2)	0.47 (0.44, 0.51)
Unknown	834	0.06 (0.05, 0.08)	0.27 (0.24, 0.3)
White	7920	0.18 (0.17, 0.18)	0.48 (0.47, 0.49)

Continued on next page

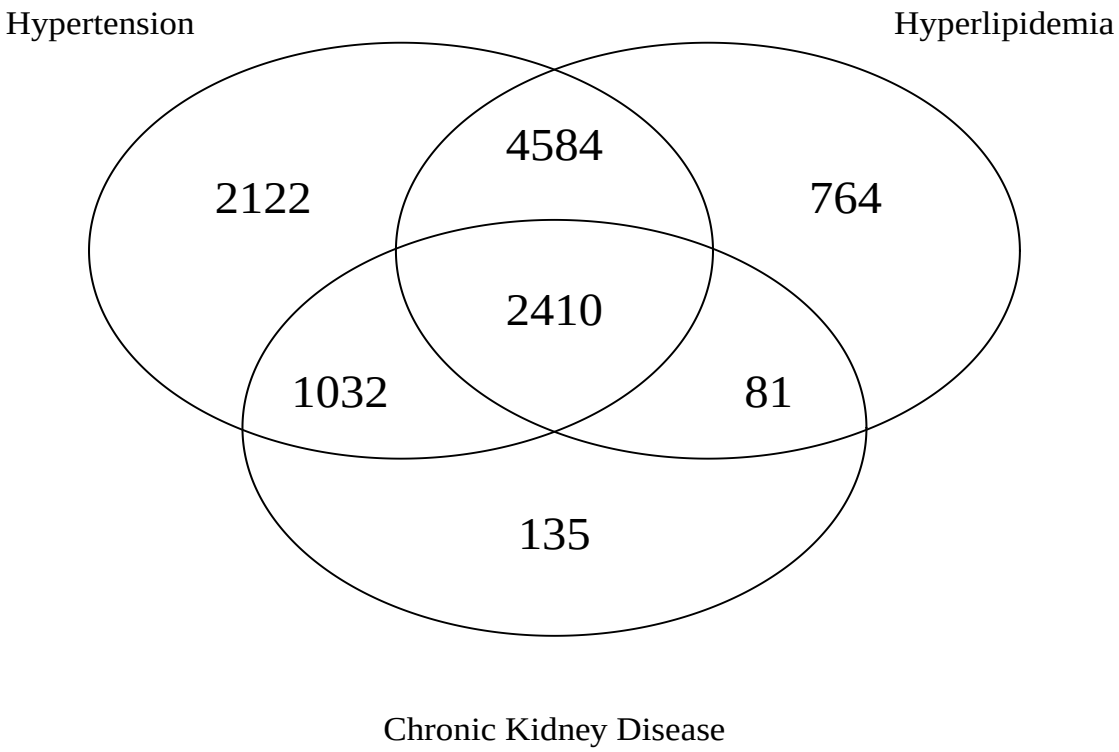
Table 2.1: Summary of categorical covariates

		Readmission Probabilities (95% CI)	
Variable	n	30-day	All-time
Index Admission Type			
Direct Emergency	400	0.2 (0.16, 0.24)	0.62 (0.58, 0.67)
Direct Observation	36	0.17 (0.04, 0.29)	0.47 (0.31, 0.64)
Elective	274	0.2 (0.15, 0.25)	0.5 (0.44, 0.56)
EU Observation	32	0.19 (0.05, 0.32)	0.53 (0.36, 0.7)
EW Emergency	3321	0.18 (0.16, 0.19)	0.5 (0.48, 0.52)
Observation Admit	5574	0.16 (0.15, 0.17)	0.48 (0.47, 0.49)
Surgical Same Day Admission	1641	0.15 (0.13, 0.17)	0.45 (0.43, 0.48)
Urgent	1807	0.21 (0.19, 0.23)	0.47 (0.45, 0.49)
Hypertension			
0	2233	0.17 (0.15, 0.18)	0.46 (0.44, 0.48)
1	10852	0.17 (0.16, 0.18)	0.49 (0.48, 0.5)
Hyperlipidemia			
0	4725	0.18 (0.17, 0.19)	0.5 (0.48, 0.51)
1	8360	0.16 (0.16, 0.17)	0.48 (0.47, 0.49)
Chronic Kidney Disease			
0	9176	0.16 (0.15, 0.17)	0.45 (0.44, 0.46)
1	3909	0.2 (0.19, 0.21)	0.56 (0.55, 0.58)

Table 2.2: Summary of numerical covariates

		Median (1st, 3rd quantiles)	
Variable	Unit	30-day	All-time
Age	year	69 (60, 77)	69 (60, 78)
Length of Stay	day	6 (3, 10)	5 (3, 8)
Glucose at Index Hospitalization	mg/dL	145 (110, 194)	145 (112, 195)
A1C at Index Hospitalization	%	10.7 (9.2, 12.2)	11.1 (9.6, 12.6)

Figure 2.2: Number of patients with one, two, or all three of the comorbidities



2.6 Logistic Regression

2.6.1 Formulation

Treating the readmission status as the outcome, we can classify patients into two classes, readmitted and readmission-free. We use logistic regression to model the relationship between the covariates and the outcome.

Let Y_i be the readmission status of the i-th patient. Let $Y_i = 1$ if the i-th patient was readmitted, and $Y_i = 0$ otherwise. We further define X_i as a vector of length $p + 1$. The first element of X_i is 1, and the rest p elements are the covariates corresponding to the i-th patient. Then, we can model the probability $P(Y_i = 1 \mid X_i)$, the readmission probability of the i-th patient given a set of covariates X_i. In a linear regression setting, the model would take the form

$$P(Y_i = 1 \mid X_i) = \beta^T X_i,$$

where β is a vector of coefficients with length $p+1$. Namely, $\beta = (\beta_0, \beta_1, \beta_2, ..., \beta_p)$. However, predictions outside the range $(0, 1)$ are not sensible in this case, since a valid probability must fall between 0 and 1. The sigmoid function solves this problem, and it takes the form

$$P(Y_i = 1 \mid X_i) = \frac{\exp(\beta^T X_i)}{1 + \exp(\beta^T X_i)}. \tag{2.1}$$

Notice that equation (2.1) can also be written as

$$\log\left(\frac{P(Y_i = 1 \mid X_i)}{1 - P(Y_i = 1 \mid X_i)}\right) = \beta^T X_i. \tag{2.2}$$

The left hand side of equation (2.2) is referred to as the log odds.

2.6.2 Interpretation of Parameters

The interpretations of model parameters vary slightly depending on the data type of covariates. Consider a logistic regression model with a single binary covariate x_1 denoting patients' sex. Let $x_1 = 0$ if the patient is female, and $x_1 = 1$ if the patient is male. Then,

for two patients i and j who are male and female respectively, $P(Y_i = 1 \mid x_1 = 1)$ represents the readmission probability of the i-th patient, given that the patient is male. Similarly, $P(Y_j = 0 \mid x_1 = 0)$ represents the readmission probability of the j-th patient, given that the patient is female. Taking the log odds for the male patient, we get

$$\log\left(\frac{P(Y_i = 1 \mid x_1 = 1)}{1 - P(Y_i = 1 \mid x_1 = 1)}\right) = \beta_0 + \beta_1 x_1 = \beta_0 + \beta_1. \tag{2.3}$$

Similarly, for the female patient, the log odds is

$$\log\left(\frac{P(Y_j = 1 \mid x_1 = 0)}{1 - P(Y_j = 1 \mid x_1 = 0)}\right) = \beta_0 + \beta_1 x_1 = \beta_0. \tag{2.4}$$

Subtracting the two log odds in (2.3) and (2.4), we get the coefficient β_1. The coefficient β_1 can therefore be interpreted as the difference in log odds between male and female patients. Alternatively, we can use odds ratio to interpret the coefficients. In the context of our example, odds ratio (OR) is the odds that a male patient will be readmitted, compared to the readmission odds of a female patient. Namely,

$$OR = \frac{P(Y_i = 1 \mid x_1 = 1)/(1 - P(Y_i = 1 \mid x_1 = 1))}{P(Y_j = 1 \mid x_1 = 0)/(1 - P(Y_j = 1 \mid x_1 = 0))} = \frac{\exp(\beta_0 + \beta_1)}{\exp(\beta_0)} = \exp(\beta_1).$$

Therefore, the readmission odds of a male patient is $\exp(\beta_1)$ times the readmission odds of a female patient. Note that in this example, the reference group is female patients, and the reference group has odds ratio 1.

To interpret the coefficient of a categorical covariate with more than two groups, we assign the group with the most observations as the reference group. For instance, the insurance covariate has three groups, "Medicare," "Medicaid," and "Other." Suppose we use x_1, x_2, and x_3 to denote these three groups respectively, the model becomes

$$P(Y_i = 1 \mid x_1, x_2, x_3) = \frac{\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3)}{1 + \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3)}.$$

We assign "Medicare" as the reference group since it has the majority of observations. Then, the "Medicare" group will have odds ratio 1. Namely, $\beta_1 = 0$. Therefore, the readmission

odds ratio of a Medicaid beneficiary to a Medicare beneficiary is

$$\frac{P(Y_i = 1 \mid x_1 = 0, x_2 = 1, x_3 = 0)/(1 - P(Y_i = 1 \mid x_1 = 0, x_2 = 1, x_3 = 0))}{P(Y_j = 1 \mid x_1 = 1, x_2 = 0, x_3 = 0)/(1 - P(Y_j = 1 \mid x_1 = 1, x_2 = 0, x_3 = 0))} = \frac{\exp(\beta_2)}{\exp(\beta_1)}$$
$$= \exp(\beta_2).$$

Thus, we conclude the readmission odds of a Medicaid beneficiary is $\exp(\beta_2)$ times the readmission odds of a Medicare beneficiary.

Continuous covariates are usually standardized so that each unit change yields meaningful interpretations. In our data, patients' glucose level at the index hospitalization is one of the continuous covariates. We standardize it so that each unit is equal to one standard deviation, $\sigma = 85.29$ mg/dL. Consider a model with glucose level x_1 as the only covariate. Then, the readmission odds ratio of a patient with glucose level $(m+1)\sigma$ mg/dL to a patient with glucose level $m\sigma$ mg/dL is

$$OR = \frac{\exp\left[\beta_0 + \beta_1(m+1)\right]}{\exp(\beta_0 + \beta_1 m)} = \exp(\beta_1).$$

Hence, one standard deviation increment in glucose is associated with $\exp(\beta_1)$ times increase in the readmission odds.

In univariate settings, the odds ratios are unadjusted for any confounders. While in multivariate settings, the odds ratio of one covariate is adjusted on all other covariates, and interpretations of coefficients are based on holding all other covariates as constants. For example, when we compare readmission odds between female and male patients, age can be a potential confounder. We fit a logistic regression model on both sex and age. Then, the odds ratio corresponding to sex is adjusted on patients having the same age. The odds ratio in such multivariate setting is known as the adjusted odds ratio.

Since we transformed the coefficient estimates into an exponentiated scale, we can quantify uncertainty of the estimates on the same exponentiated scale. Namely, the $(1-\alpha)$ level two-sided confidence interval for β_j is

$$\exp[\hat{\beta}_j \pm z_{1-\alpha/2} \cdot se(\hat{\beta}_j)].$$

Note that logistic regression has no closed-form solution for the maximum likelihood estimator, numerical methods are used to estimate its coefficients. We fit logistic regression models in R using the `glm` function (see Appendix A for details on R code).

2.7 Survival Analysis Methods

One limitation of logistic regression is that its response is binary. Thus, information on patients' time to readmission is not incorporated in the model. Alternative to modeling the readmission status as a binary outcome, we can consider time-to-readmission as the outcome. Many methods in survival analysis model the time to a certain event such as death. In this section, we define hospital readmission as the event of interest and discuss survival analysis methods in modeling time-to-readmission.

2.7.1 Survival Functions and the Hazard

In survival analysis, we first need to define the initiating event and the terminating event. In the context of hospital readmission, the initing event is the discharge from the index hospitalization, and the terminating event is readmission. One quantity of interest in survival analysis is the probability of survival up to a given time point t. Let T be a random variable that denotes this quantity of interest. Mathematically, we write the probability function as

$$S(t) = P(T > t).$$

The function $S(t)$ is referred to as the survival function. In our context, survival function is the probability of readmission-free up to time t. Another quantity of interest is the hazard, which is defined as

$$h(t) = \lim_{\delta \to 0} \frac{P(t \leq T < t + \delta \mid T \geq t)}{\delta}.$$

In our example, the hazard can be interpreted as the instantaneous readmission rate at time t.

By definition, $S(t) = 1 - F(t)$, where $F(t)$ is the cumulative distribution function of the random variable T. Therefore, we can write the probability density function of T as

$$f(t) = -\frac{d}{dt}S(t). \tag{2.5}$$

And by the conditional probability,

$$h(t) = \lim_{\delta \to 0} \frac{P(t \leq T < t + \delta \mid T \geq t)}{\delta} = \lim_{\delta \to 0} \frac{P(t \leq T < t + \delta)}{P(T \geq t)\delta} = \frac{f(t)}{S(t)}. \tag{2.6}$$

From equations (2.5) and (2.6), it can also be shown that

$$h(t) = -\frac{d}{dt} \log S(t). \tag{2.7}$$

Lastly, we define the cumulative hazard as $H(t) = \int_0^t h(u)\ du$.

2.7.2 Cox Proportional-Hazards Model

The Cox Proportional Hazards (PH) model assumes that the hazard functions of every subgroup are constant and proportional to each other. Suppose the hazard functions of two subgroups of individuals are $h_0(t)$ and $h_1(t)$. Then, in a PH model, $h_1(t) = c \cdot h_0(t)$, where c is a constant. The PH model can be formulated as

$$h(t \mid X) = h_0(t) \exp(\beta^T X).$$

The hazard $h_0(t)$ is also referred to as the baseline hazard. The interpretation of parameters in a PH model is similar to that in a logistic regression model, except we consider the hazard ratio instead of the odds ratio. For example, consider a model with patient sex x_1 and hypertensive x_2 as covariates, we can express the model as

$$h(t \mid x_1, x_2) = h_0(t) \exp(\beta_1 x_1 + \beta_2 x_2).$$

Comparing the readmission hazards of male hypertensive patients to female hypertensive patients, we have

$$\frac{h(t \mid x_1 = 1, x_2 = 1)}{h(t \mid x_1 = 0, x_2 = 1)} = \frac{h_0(t) \exp(\beta_1 \cdot 1 + \beta_2 \cdot 1)}{h_0(t) \exp(\beta_1 \cdot 0 + \beta_2 \cdot 1)} = \exp(\beta_1). \tag{2.8}$$

We can conclude that the readmission hazard of male hypertensive patients is $\exp(\beta_1)$ times the readmission hazard of female hypertensive patients.

The coefficients in a Proportional-Hazards model can be estimated using the partial likelihood function introduced by Cox (1975). The partial likelihood function does not depend on the baseline hazard $h_0(t)$.

2.7.3 Accelerated Failure Time Model

The accelerated failure time (AFT) model states that the an individual with covariate X at time t has the same survival function as an individual with baseline survival function at time $\exp(\beta^T X)t$, where $\beta = (\beta_1, \beta_2, \ldots, \beta_p) \in \mathbb{R}^p$ (Klein & Moeschberger, 2006). Formally,

$$S(t \mid X) = S_0\left[\exp(\beta^T X)t\right]. \tag{2.9}$$

The key assumption of an AFT model is that the survival time of one group is a constant multiple of the survival time of the baseline group. In equation (2.9), the constant term $\exp(\beta^T X)$ is referred to as the acceleration factor, and it changes the time scale of the baseline survival function. To derive the hazard function of an AFT model, we first consider its density function. By equation (2.5), the density function is

$$f(t \mid X) = -\frac{d}{dt}S(t \mid X) = -\frac{d}{dt}S_0\left[\exp(\beta^T X)t\right] = -\exp(\beta^T X) \cdot \frac{d}{dt}S_0\left[\exp(\beta^T X)t\right].$$

Because the density function of the baseline is

$$f_0(t) = -\frac{d}{dt}S_0(t),$$

we can write the density function as

$$f(t \mid X) = \exp(\beta^T X) f_0\left[\exp(\beta^T X)t\right].$$

Therefore, the hazard function becomes

$$h(t \mid X) = \frac{f(t \mid X)}{S(t \mid X)} = \frac{\exp(\beta^T X) f_0\left[\exp(\beta^T X)t\right]}{S_0\left[\exp(\beta^T X)t\right]} = \exp(\beta^T X) \cdot h_0\left[\exp(\beta^T X)t\right].$$

Unlike the interpretations of a Proportional-Hazards model parameters which are based on hazard ratios, AFT models can be interpreted as the mean or median time-to-event ratios. Consider an AFT model with one binary covariate x_1 denoting an individual's sex. Then, taking the mean time-to-readmission ratio of a male to a female patient, we get

$$\frac{\mathbb{E}[T \mid x_1 = 1]}{\mathbb{E}[T \mid x_1 = 0]} = \frac{\exp(-\beta_1) \cdot \mathbb{E}[T \mid x_1 = 0]}{\mathbb{E}[T \mid x_1 = 0]} = \exp(-\beta_1).$$

And we conclude that the mean time-to-readmission of a male patient is $\exp(-\beta_1)$ times that of a female patient. Alternatively, if we model the acceleration factor as $\exp(-\beta^T X)$, then the mean time ratio would be $\exp(\beta_1)$.

2.7.4 Weibull Distribution

The baseline hazard function of an AFT model can be estimated using parametric models. A common choice of the baseline hazard is the Weibull model. The survival function of a Weibull distribution is

$$S(t) = \exp\left[-(\lambda t)^{\alpha}\right]$$

with $\lambda, a > 0$. By equation (2.7), the hazard function can be written as

$$h(t) = \alpha \lambda^{\alpha} t^{\alpha - 1}.$$

If we use a Weibull distribution to model the baseline survival function in an AFT model, then

$$S(t \mid X) = S_0\left[\exp(-\beta^T X)t\right] = \exp\left\{-\left[\lambda \exp(-\beta^T X)t\right]^{\alpha}\right\}.$$

Let $\lambda = \exp(-\beta_0)$, we can rewrite the model as

$$S(t \mid X) = \exp\left\{-\left[\exp(-\tilde{\beta}^T \tilde{X})t\right]^{\alpha}\right\},$$

where $\tilde{\beta} = (\beta_0, \beta_1, \ldots, \beta_p) \in \mathbb{R}^{p+1}$, and $\tilde{X} = (1, X_1, X_2, \ldots, X_p) \in \mathbb{R}^{p+1}$. With a Weibull baseline hazard, an AFT model with one binary covariate X_1 is

$$S(t \mid X_1 = x) = \exp\left\{-\left[\exp(-\beta_0 - \beta_1 x)t\right]^{\alpha}\right\} = \exp\left[-\exp(-\beta_0 \alpha - \beta_1 x \alpha)t^{\alpha}\right].$$

By equation (2.5),

$$f(t \mid X_1 = x) = \exp(-\beta_0\alpha - \beta_1 x\alpha)\alpha t^{\alpha-1} \exp\left[-\exp(-\beta_0\alpha - \beta_1 x\alpha)t^{\alpha}\right].$$

From (2.6), we can show that

$$h(t \mid X_1 = x) = \exp(-\beta_0\alpha)\alpha t^{\alpha-1} \cdot \exp(-\beta_1 x\alpha).$$

Note that if we let $h_0(t) = \exp(-\beta_0\alpha)\alpha t^{\alpha-1}$ and $c = \exp(-\beta_1 x\alpha)$, then the AFT model with a Weibull distribution as the baseline survival function would take the form $h(t) = c \cdot h_0(t)$ which also satisfies proportional hazards.

2.8 Model Diagnostics

One assumption of the proportional-hazards model, as its name suggests, is that the hazard functions of every subgroup are proportional to each other. Grambsch and Therneau (1994) suggested a regression approach to test for non-proportionality in hazards by adding a time-dependent coefficient to the model. They showed that many tests for proportional hazards are equivalent to testing for non-zero slopes in a generalized linear regression of the scaled Schoenfeld residuals on a function of time. Under the proportional hazards assumption, smoothing the scaled Schoenfeld residual plot can be used to visualize the time-varying effect of hazards. As an example, in a proportional hazards model with patient sex X_1 as a single covariate, we create a new time-dependent covariate $X_2 = X_1 \cdot g(t)$, where $g(t)$ can be an arbitrary function of time. Therefore, the hazard becomes

$$h(t \mid X_1, X_2) = h_0(t)\exp(\beta_1 X_1 + \beta_2 X_2) = h_0(t)\exp\left[\beta_1 X_1 + \beta_2 \left(X_1 \cdot g(t)\right)\right].$$

Comparing the readmission hazard of male patients to female patients, we get

$$\frac{h(t \mid 1)}{h(t \mid 0)} = \exp\left[\beta_1 + \beta_2 g(t)\right]. \tag{2.10}$$

Notice that in (2.10), the hazard ratio depends on time t when $\beta_2 \neq 0$. Therefore, the null hypothesis that $\beta_2 = 0$ can be used to test the proportional hazards assumption. Grambsch

and Therneau (1994) also showed that the functional form of the time-varying coefficient can be visualized by smoothing a scaled Schoenfeld residual plot. Namely, the scaled Schoenfeld residual plot gives us an estimate of $\tilde{\beta}(t)$, where $\tilde{\beta}(t) = \beta_1 + \beta_2 g(t)$. Therefore, the proportional hazards assumption holds when the estimate of $\tilde{\beta}(t)$ is a horizontal line. Plots showing the time-varying effect of readmission hazards for various risk factors are available in the results section (Figure 3.1-3.3).

To assess the fit of the AFT model, we can examine the deviance residual plots (Klein & Moeschberger, 2006). The deviance residual for the i-th observation takes the form

$$D_i = \text{sign}(M_i)\sqrt{-2[M_i + \delta_i \log(\delta_i - M_i)]},$$

where $\delta_i = 1$ if the i-th patient was readmitted, otherwise $\delta_i = 0$. The Martingale residual M_i for the i-th observation is defined as $M_i = \delta_i - \hat{H}(T_i \mid X_i)$. It can also be shown that the estimated cumulative hazard $\hat{H}(T_i \mid X_i)$ is an estimate of $\mathbb{E}[\delta_i]$. Therefore, a positive deviance residual indicates that the actual event happened sooner than the predicted event time; and a negative deviance residual suggests that the actual event happened later than the predicted time. The deviance residuals plots for the multivariate AFT model were shown in Appendix B.

Chapter 3

RESULTS

In this chapter, we first study the difference of risk factors over various time periods after patients were discharged from index hospitalization. Then, we employ the Cox PH model on each covariate and compare the results with an AFT model. Next, we discuss the time-varying effect which can lead to non-proportional hazards. Finally, we present a multivariate AFT model and discuss the interpretations of parameter estimates.

3.1 Multiple Logistic Regression on Different Time Intervals

To compare statistically significant risk factors over time, we consider three different readmission time intervals: readmitted within 30 days, between 30 and 90 days, and between 90 and 180 days. Using methods described in section 2.6, we fit a logistic regression model on each time interval with the outcome being the readmission status (1 being readmitted; 0 being readmission-free within the corresponding time interval). Table 3.1 shows the resulting adjusted odds ratios with 95% confidence intervals.

According to table 3.1, patient sex was not a statistically significant risk factor for readmission. Patients with "EW emergency" had a higher 30-day readmission odds than patients with "Observation Admit" admission type ($p = 0.042$). Patients admitted as "Urgent" also had a higher 30-day readmission odds ($p < 0.001$). Medicaid patients had a higher 30-day readmission odds ($p = 0.044$), but a lower 30 to 90-day readmission odds than Medicare patients ($p = 0.026$). Black/African American patients were more likely to be readmitted compared to White patients for the 90-180 days period ($p = 0.009$), but the difference in readmission odds was not statistically significant for earlier time periods. For all three time intervals, patients with chronic kidney disease had a higher readmission odds than patients

without. The length of index hospitalization was also shown to be significant across all three time intervals. In particular, one day increment in the length of stay was associated with a 5% increase in 30-day readmission odds ($p < 0.001$), a 2% increase in the 30 to 90-day odds ($p = 0.003$), but a 6% drop in 90 to 180-day odds ($p < 0.001$). Age and glucose were not significant predictors in all three logistic regression models. However, one standard deviation increment in A1C was correlated with a 25 percent decrease in the 30-day readmission likelihood ($p < 0.001$). This relationship might be because of the intensification of glucose therapy for patients with higher A1C, as Wei et al. (2013) found in their study that glucose therapy intensification was associated with a significantly lower early readmission risk.

Table 3.1: Adjusted odds ratios for readmission within 30, 30-90, and 90-180 days with 95% confidence interval

	Adjusted Odds Ratio (95% CI)		
Variable	30 days	30-90 days	90-180 days
Sex			
Female	1 (reference)	1 (reference)	1 (reference)
Male	1.05 (0.95, 1.16)	1 (0.89, 1.13)	0.92 (0.79, 1.07)
Index Admission Type			
Observation Admit	1 (reference)	1 (reference)	1 (reference)
Direct Emergency	1.4 (1.07, 1.82)	1.62 (1.2, 2.18)	1.05 (0.69, 1.59)
Direct Observation	1.05 (0.4, 2.77)	1.14 (0.34, 3.82)	1.76 (0.6, 5.15)
Elective	1.53 (1.11, 2.09)	0.81 (0.51, 1.28)	1.07 (0.65, 1.76)
EU Observation	0.98 (0.33, 2.9)	0.42 (0.06, 3.15)	0.94 (0.22, 4.03)
EW Emergency	1.13 (1, 1.27)*	1.09 (0.94, 1.25)	0.93 (0.78, 1.11)
Surgical Same Day Admission	1.01 (0.85, 1.19)	0.74 (0.59, 0.92)*	0.76 (0.58, 0.99)*
Urgent	1.53 (1.32, 1.77)*	1.04 (0.86, 1.26)	0.81 (0.63, 1.05)

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Table 3.1: Adjusted odds ratios for readmission within 30, 30-90, and 90-180 days with 95% confidence interval

	Adjusted Odds Ratio (95% CI)		
Variable	30 days	30-90 days	90-180 days
Insurance			
Medicare	1 (reference)	1 (reference)	1 (reference)
Medicaid	1.27 (1.01, 1.59)*	0.68 (0.49, 0.96)*	0.95 (0.65, 1.37)
Other	1 (0.9, 1.11)	0.94 (0.83, 1.07)	0.97 (0.83, 1.14)
Ethnicity			
White	1 (reference)	1 (reference)	1 (reference)
American Indian/Alaska Native	1.72 (0.86, 3.44)	1.38 (0.57, 3.3)	0.76 (0.18, 3.17)
Asian	1.09 (0.85, 1.38)	0.95 (0.69, 1.3)	1.05 (0.72, 1.52)
Black/African American	1.04 (0.92, 1.19)	1.04 (0.89, 1.22)	1.28 (1.06, 1.54)*
Hispanic/Latino	1.1 (0.9, 1.36)	1.2 (0.93, 1.55)	1.21 (0.89, 1.65)
Other	0.98 (0.79, 1.21)	1.07 (0.82, 1.38)	0.98 (0.7, 1.37)
Comorbidities			
Hypertension	1.03 (0.9, 1.18)	0.97 (0.82, 1.16)	0.87 (0.71, 1.07)
Hyperlipidemia	0.93 (0.84, 1.03)	0.92 (0.81, 1.04)	0.86 (0.74, 1)
Chronic Kidney Disease	1.18 (1.06, 1.32)*	1.21 (1.05, 1.38)*	1.37 (1.16, 1.62)*
Continuous Variables			
Length of Stay (per $\sigma = 8$ days)	1.23 (1.18, 1.29)*	1.08 (1.03, 1.14)*	0.79 (0.71, 0.88)*
Age (per 10 years)	0.97 (0.93, 1.01)	1 (0.95, 1.05)	1 (0.93, 1.06)
Glucose (per $\sigma = 85.29$ mg/dL)	0.99 (0.94, 1.04)	0.97 (0.91, 1.03)	1.04 (0.97, 1.12)
A1C (per $\sigma = 2.14\%$)	0.85 (0.81, 0.9)*	0.87 (0.82, 0.93)*	0.94 (0.86, 1.01)

3.2 Univariate Cox PH and AFT Models

In this section, we fit the Cox Proportional Hazards and the accelerated failure time models discussed in section 2.7 by considering time-to-readmission as the outcome of interest. Then, we compare the results from both models.

Table 3.2: Hazard ratios and mean/median readmission time ratios with 95% confidence intervals

	Cox PH Model	AFT Model
Covariate	Hazard Ratios	Mean/Median Readmission Time Ratios
Sex		
Female	1 (reference)	1 (reference)
Male	0.95 (0.91, 1)	1.15 (1.00, 1.32)
Index Admission Type		
Observation Admit	1 (reference)	1 (reference)
Direct Emergency	1.49 (1.31, 1.71)*	0.32 (0.22, 0.45)*
Direct Observation	0.98 (0.61, 1.57)	1.07 (0.3, 3.78)
Elective	1.11 (0.93, 1.33)	0.77 (0.48, 1.22)
EU Observation	1.18 (0.73, 1.9)	0.63 (0.18, 2.24)
EW Emergency	1.1 (1.03, 1.17)*	0.79 (0.67, 0.93)*
Surgical Same Day Admission	0.89 (0.82, 0.97)*	1.38 (1.11, 1.72)*
Urgent	1.16 (1.07, 1.25)*	0.71 (0.57, 0.87)*

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Table 3.2: Hazard ratios and mean/median readmission time ratios with 95% confidence intervals

	Cox PH Model	AFT Model
Covariate	Hazard Ratios	Mean/Median Readmission Time Ratios
Insurance		
Medicare	1 (reference)	1 (reference)
Medicaid	1.05 (0.93, 1.19)	0.88 (0.64, 1.22)
Other	0.93 (0.89, 0.98)*	1.22 (1.07, 1.4)*
Ethnicity		
White	1 (reference)	1 (reference)
American Indian/Alaska Native	1.84 (1.27, 2.67)*	0.18 (0.07, 0.48)*
Asian	1.06 (0.93, 1.2)	0.85 (0.61, 1.2)
Black/African American	1.22 (1.14, 1.3)*	0.56 (0.47, 0.66)*
Hispanic/Latino	1.21 (1.1, 1.35)*	0.57 (0.43, 0.75)*
Other	0.99 (0.88, 1.11)	1.03 (0.77, 1.4)
Comorbidities		
Hypertension	1.09 (1.01, 1.16)*	0.79 (0.66, 0.94)*
Hyperlipidemia	0.93 (0.88, 0.98)*	1.22 (1.06, 1.4)*
Chronic Kidney Disease	1.39 (1.31, 1.46)*	0.4 (0.34, 0.46)*
Continuous Covariates		
Length of Stay (per $\sigma = 8$ days)	1.1 (1.08, 1.13)*	0.78 (0.74, 0.83)*
Age (per 10 years)	1 (0.98, 1.02)	0.99 (0.94, 1.04)
Glucose (per $\sigma = 85.29$ mg/dL)	0.98 (0.96, 1.01)	1.05 (0.98, 1.12)
A1C (per $\sigma = 2.14\%$)	0.87 (0.85, 0.89)*	1.47 (1.37, 1.57)*

As shown in Table 3.2, statistically significant covariates based on the two models were marked using asterisks. Patients with emergency related admission types had higher readmission hazard. Medicare/Medicaid beneficiaries had higher hazard than patients with other insurance types. American Indian/Alaska Native, Black/African American, and Hispanic/Latino had relatively higher readmission hazard. Patients with Hyperlipidemia had lower readmission hazard than patients without, which may be because a number of patients in our population who were diagnosed with hypertension and chronic kidney disease did not have hyperlipidemia. Together with the results from the previous logistic regression models, we saw that having chronic kidney disease was the most significant risk factor among those three common comorbidities. The findings from the Cox PH and the AFT model agree qualitatively. In particular, comparing the two models, we found no difference in the direction of association between covariates and the outcome. Note that for results to agree qualitatively, a hazard ratio greater than 1 must have a mean/median readmission time ratio less than 1, and vice versa.

3.3 The Time-varying Effect of Non-proportional Hazard Ratios

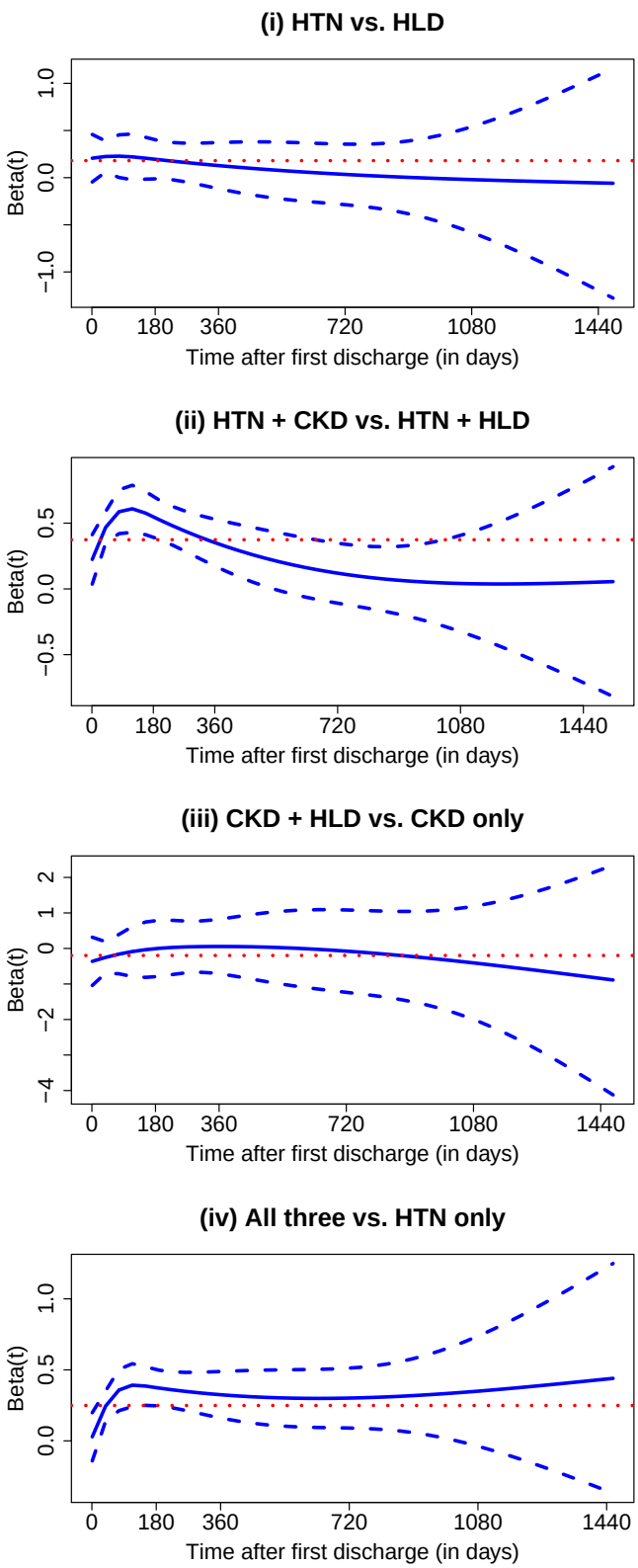
Since the Cox PH model assumes constant hazard ratios for all groups, we need to check the proportional hazards assumption. Using the method described in section 2.8, we conduct a hypothesis test on each covariate. The resulting p-values are listed in Table 3.3. Under $\alpha = 0.05$, we conclude that for most covariates, the proportional hazards assumption fails to hold. Non-proportionality may be due to a number of reasons. For instance, continuous covariates may have nonlinear effects, leading to non-proportional hazards. Here, we consider the time-varying effects of covariates. Namely, we study the behavior of hazard ratios as a function of time.

Table 3.3: p-values from proportional hazards tests, i.e. score tests for slope = 0

Covariate	p-value
Sex	0.001
Admission Type	< 0.001
Insurance	0.159
Ethnicity	< 0.001
Hypertension	0.093
Hyperlipidemia	0.015
Chronic Kidney Disease	0.015
Length of Stay (per $\sigma = 8$ days)	< 0.001
Age (per 10 years)	0.492
Glucose (per $\sigma = 85.29$ mg/dL)	0.917
A1C (per $\sigma = 2.14\%$)	0.001

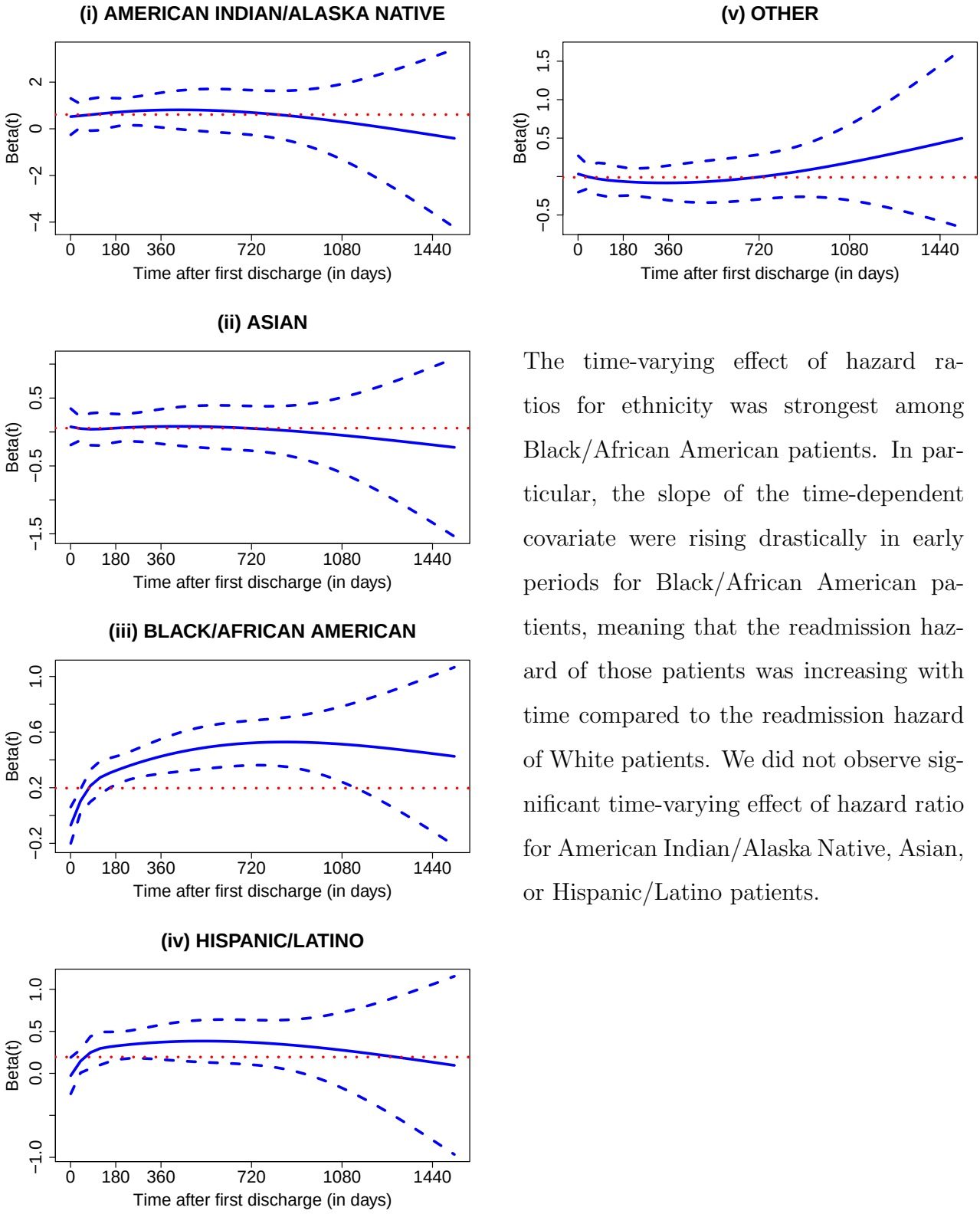
To visualize the time-varying effect of hazard ratios, we constructed Figures 3.1-3.6 using the method described in section 2.8.

Figure 3.1: Time-varying effect of hazard ratios: comorbidity



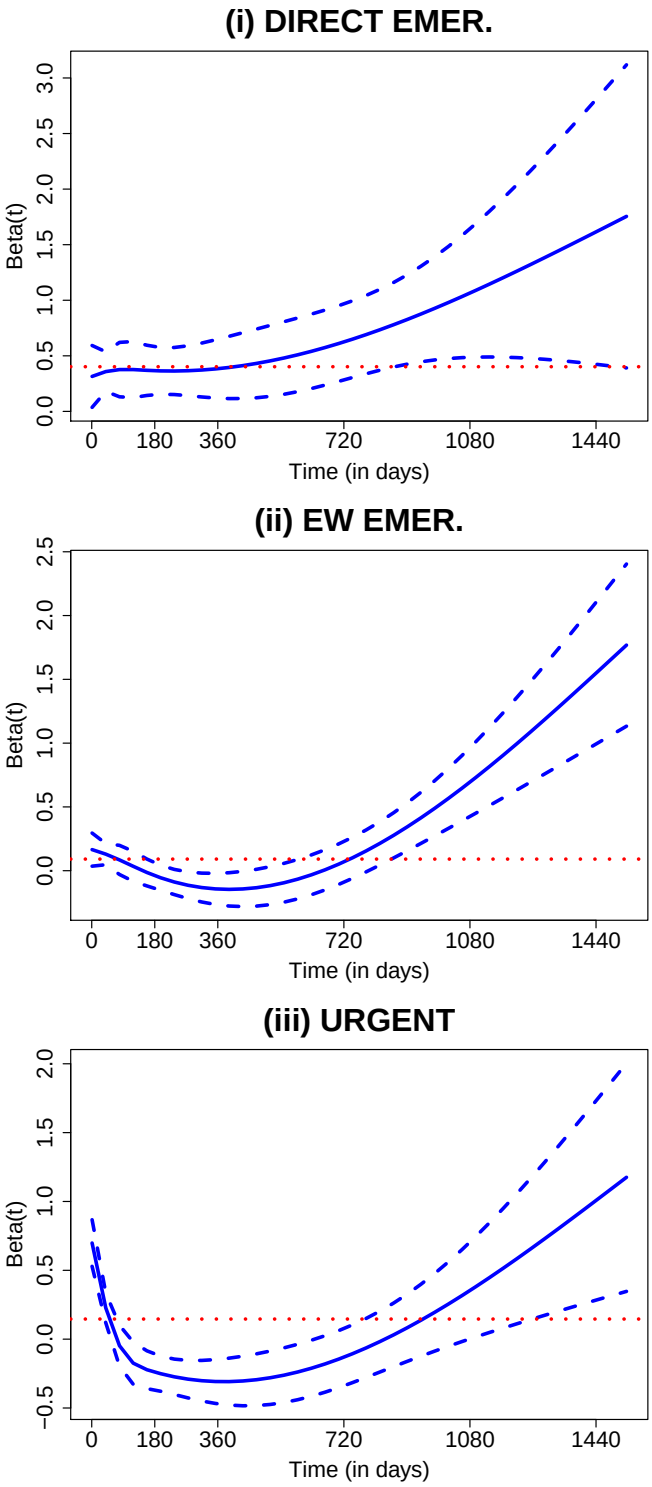
The red dotted lines are the estimated coefficients from the Cox PH model. Note that hazard ratios can be obtained by exponentiating the coefficients. Therefore, the results would be the same if we change the y-axis to the hazard ratio. The blue lines are the scaled Shoenfeld residuals smoothed using regression spline. The blue dashed lines are the confidence intervals obtained from the smoothing. Plot (i) showed no strong pattern of non-proportionality, as the confidence interval bands covered the coefficient estimate entirely. From plot (ii), we observed that the slope of the time-dependent coefficient had a substantial increase within around 180 days after discharge from index hospitalization, indicating that among hypertensive patients, the early readmission hazard ratio of patients with chronic kidney disease to patients without increased with time. Amongst all three comorbidities, chronic kidney disease had the strongest time-varying effect of readmission hazard ratio.

Figure 3.2: Time-varying effect of hazard ratios: ethnicity

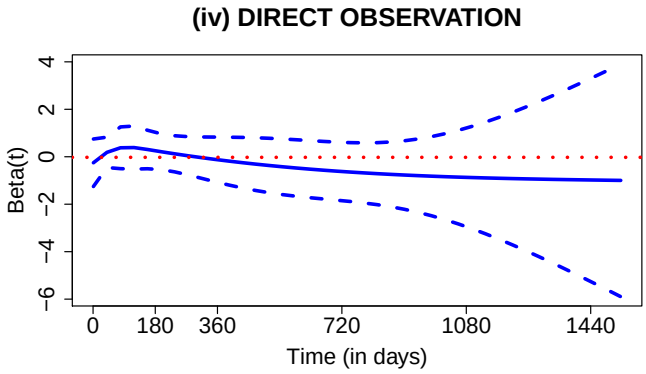


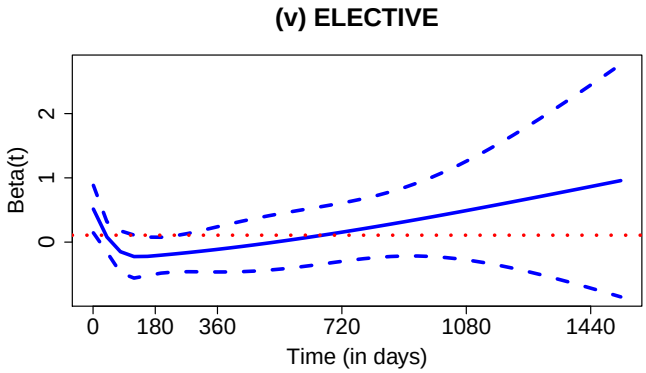
The time-varying effect of hazard ratios for ethnicity was strongest among Black/African American patients. In particular, the slope of the time-dependent covariate were rising drastically in early periods for Black/African American patients, meaning that the readmission hazard of those patients was increasing with time compared to the readmission hazard of White patients. We did not observe significant time-varying effect of hazard ratio for American Indian/Alaska Native, Asian, or Hispanic/Latino patients.

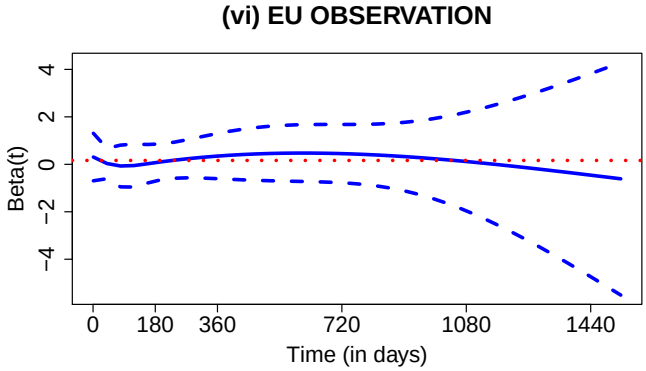
Figure 3.3: Time-varying effect of hazard ratios: admission types

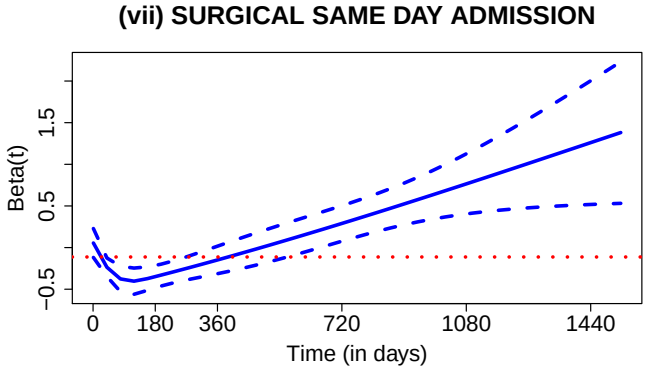


The reference group for admission types is observation admission. The time-varying patterns were similar for EW emergency, urgent, and surgical same-day admissions. The estimated time-dependent coefficients for them were U-shaped, suggesting that the readmission hazards for patients with the aforementioned admission types decreased shortly after discharge, but they all increased substantially later on. The time points that the estimated slopes turned positive vary by admission types. For EW emergency and urgent patients, the hazard ratio started to increase at around one year after discharge. For surgical same-day admission patients, the time at which hazard ratio started to increase was approximately 3 months after discharge from index hospitalization. Note that in section 3.1, we compared the adjusted readmission odds ratios for various time intervals. Here, the time-varying pattern also relates to our previous findings presented in Table 3.1.



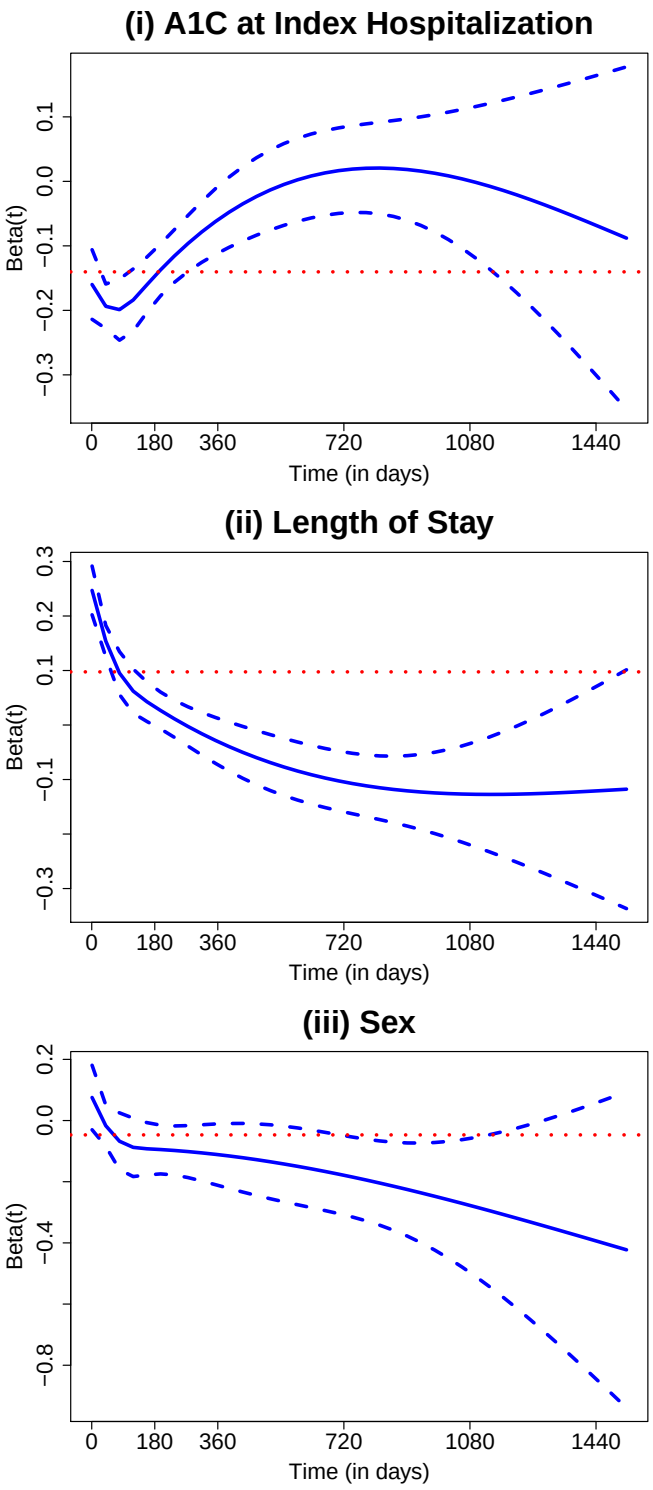






As an example, according to Table 3.1, the adjusted readmission odds ratios for surgical same-day admission patients dropped from 30-day to 180-day readmission. And in plot (vii), we also observed the same trend before 180 days.

Figure 3.4: Time-varying effect of hazard ratios: A1C, length of stay, and sex



Although the Cox PH model suggested a negative relationship between the A1C and the hazard ratio, which may be due to different levels of treatment received by patients, the estimated slope for the time-dependent covariate increases significantly after around 90 days. In other words, even though we found that an increase in A1C was associated with a decrease in the hazard ratio, when plotting the time-varying hazard ratio against time, we found that the hazard ratio was increasing. For the length of stay, the readmission hazard ratio decreased monotonically with time, except for the second half of the plot in which the confidence interval became wider due to lack of observations. Notice that the direction of association between the length of stay and the readmission hazard ratio also changes. In particular, an increase in length of stay is associated with a hazard ratio above 1 at early times after discharge but then with a hazard ratio below 1.

For the length of stay covariate, we also observed a similar pattern from the previous logistic regression results as shown in Table 3.1. The odds ratio associated with a unit increase in the length of stay for 30-day readmission was above 1, but the odds ratio was below 1 for 90 to 180-day readmission.

3.4 Multivariate AFT Model

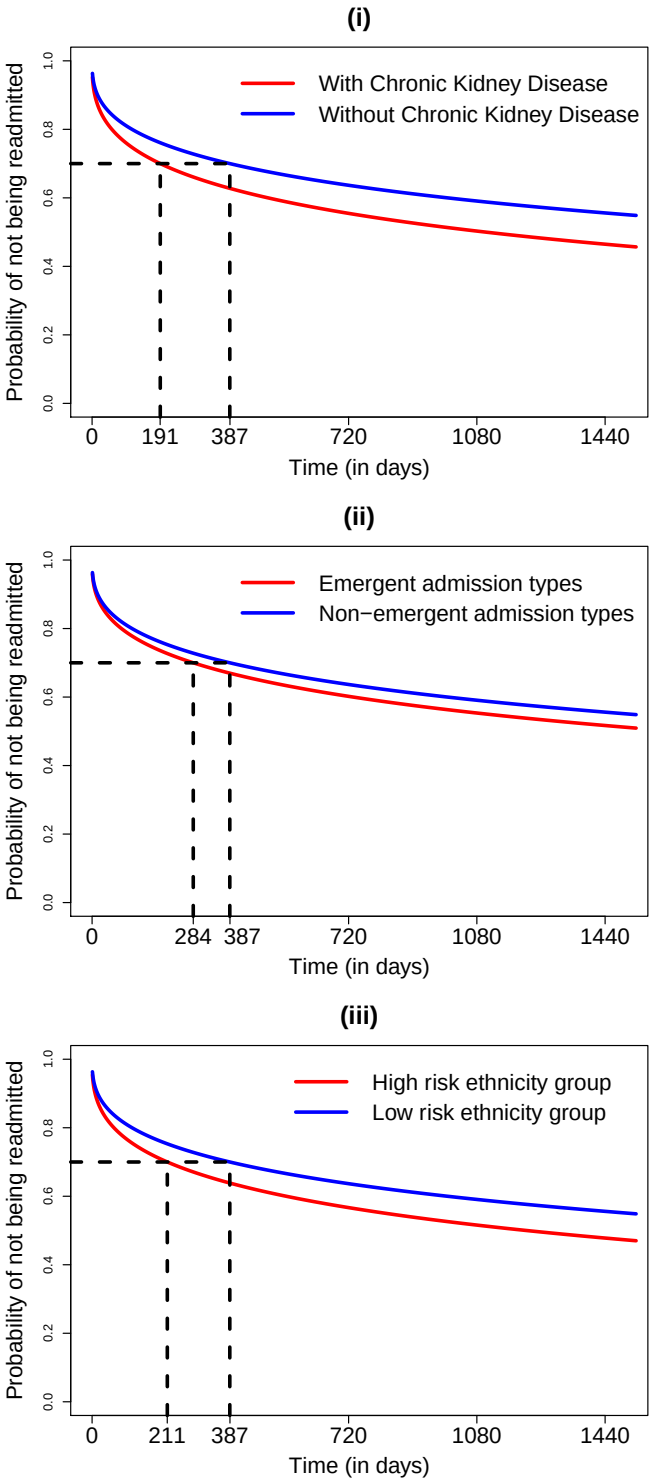
So far, we have only considered univariate Cox PH and AFT models. To build a multivariate AFT model, we first collapse subgroups of categorical covariates by constructing a new set of covariates. In particular, for admission types, we merge Direct Emergency, EW Emergency, and Urgent into one subgroup and other admission types into one subgroup. For ethnicity, we combine subgroups based on high/low readmission risks. American Indian/Alaska Native, Black/African American, and Hispanic/Latino patients are in the high readmission risk group. Lastly, we combine patients enrolled in Medicare and Medicaid into one group.

Table 3.4 shows the model coefficient estimates. Notice that the A1C at Index Hospital Stay has a coefficient greater than 1. Although one might expect patients with higher A1C to have a higher risk of readmission, our data suggests that an increase in A1C is associated with a longer mean time to readmission. Again, this might be due to differences in the level of treatment received by patients. As described in section 2.7.3, coefficients can be viewed as the mean/median readmission time ratios. Therefore, we can visualize their effects by comparing the time-to-readmission among subgroups. Each figure in 3.5 and 3.6 corresponds to one covariate in the AFT model.

Table 3.4: Mean/median readmission time ratios from multivariate AFT model with 95% confidence intervals

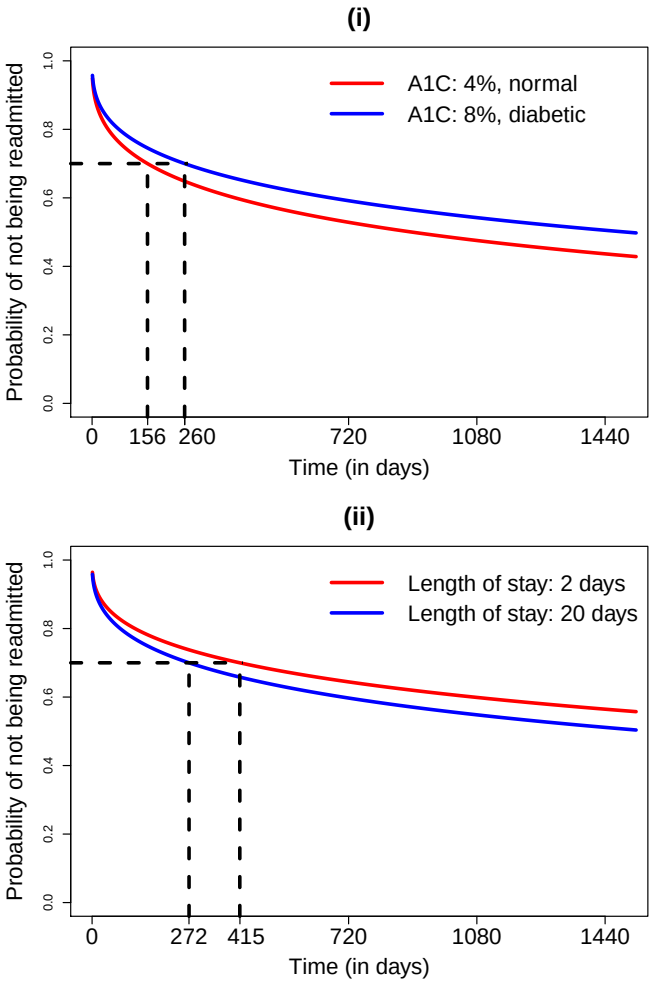
Covariate	Mean/Median Readmit Time Ratios
Chronic Kidney Disease Diagnosis	0.50 (0.43, 0.57)
A1C at Index Hospital Stay (per $\sigma = 2.14\%$)	1.32 (1.23, 1.41)
Length of Stay (per $\sigma = 8$ days)	0.83 (0.78, 0.88)
Emergent Admission Types	0.73 (0.64, 0.84)
High Readmission Risk Ethnicity Group	0.55 (0.47, 0.63)

Figure 3.5: Readmission time ratios: categorical covariates



The y-axis of each figure is the probability of not being readmitted. And the probability of readmission is simply the complement of that. As mentioned in section 2.7.3, we would normally interpret the parameters as median readmission time ratios, i.e. the readmission time ratio when the probability of readmission is 0.5. However, in our case, less than 50 percent of patients were readmitted overall. Therefore, we consider the readmission time ratio when the readmission rate is 0.3, since the AFT model assumes that the readmission time ratio between two groups of patients is always constant. According to figure (i), for patients without chronic kidney disease, the probability of readmission would be 0.3 at 387 days after discharge from index hospitalization. However, the probability of readmission would reach 0.3 as soon as at 191 days after discharge from index hospitalization. The readmission time ratio, 191/387, is the exponentiated coefficient from the AFT model.

Figure 3.6: Readmission time ratios: numerical covariates



For numerical covariates, we can choose any subgroups of patients to compare. As an example, A1C at 4% is within the normal range while 8% is considered as diabetic. Based on the first figure in 3.6, the time-to-readmission is longer among patients with a higher A1C during index hospitalization. This result might be counter-intuitive. But as mentioned before, it may be due to differences in the levels of treatment received by the patients. In the second figure of 3.6, we look at patients whose length of stay are 2 and 20 days. Among all hospitalized patients, a length of stay of 20 days is considered relatively long. And we expect patients' conditions to be more severe. The result matches our hypothesis. Patients with longer length of index hospitalization has a shorter mean/median time-to-readmission based on our model. Note that the time ratio is assumed to be the same regardless of admission types, ethnicity groups, A1C, or length of stay. For example, the readmission time ratios of patients with chronic kidney disease to patients without are the same across all admission types.

To assess the fit of the multivariate AFT model, we plotted the deviance residuals (see Appendix B). Notice that for categorical covariates, the deviance residuals were systematically positively skewed in the reference groups. And as mentioned in section 2.8, a positive deviance residual means that the actual readmission happened sooner than the predicted time-to-readmission. Therefore, for patients with lower readmission risk, the predicted time-to-readmission were longer than the actual readmission time. On the other hand, for high readmission risk patients, predictions were shorter than the actual time-to-readmission.

Chapter 4

CONCLUSION

To summarize, we have identified some statistically significant risk factors for hospital readmission of type 2 diabetes patients and applied survival analysis methods to model time-to-readmission. In section 3.1, we viewed readmission as a classification problem and compared the adjusted odds ratios among patients readmitted within 30, 30-90, and 90-180 days. The probability representation of odds ratios is easy to interpret, but logistic regression is limited in that its outcome is binary. When the outcome of interest is time-to-readmission, logistic regression cannot be used unless we dichotomize the outcome. Logistic regression also does not account for censored observations. Although we have not discussed censoring in this thesis due to lack of available data, censoring events increase the complexity of data structure and need to be carefully accommodated. As an example, in the context of hospital readmission, a right-censored observation may be a patient who had not been readmitted before the latest available date in the database. We cannot consider such observation as readmission-free because the readmission event may happen later. However, we know that the patient is readmission-free between discharge from the index hospitalization and the latest date in the database. Probabilities for both censored and uncensored observations contribute to calculating the maximum likelihood estimator for the Cox PH and the AFT models.

In this thesis, we made the following assumptions. First, because the data were collected only from the Beth Israel Deaconess Medical Center, patients readmitted to other hospital facilities were truncated due to lack of records. Moreover, between the recorded index hospitalization and the readmission event time, a patient may experience multiple readmissions at other hospitals. In such cases, the time-to-readmission would be longer than the actual

time. However, in our analyses, we assumed no such cases exist. In addition, we included patients as long as they have type 2 diabetes as one of their medical diagnoses for both the index hospitalization and the readmission, although patients may be admitted or readmitted for primary causes other than type 2 diabetes.

Models discussed in this thesis, Cox PH and AFT, also have strong assumptions. For the Cox PH model, the hazard ratios are assumed to be constant, and violation of this assumption may lead to invalid inferences. In sections 2.8 and 3.3, we showed methods for testing the proportionality assumption and graphs for visualizing the time-varying effect of hazard ratios. Compared to the Cox PH model, the AFT model relaxed the constant hazards assumption, but it assumes that the acceleration factor is constant. Research on the Cox PH and the AFT models showed that the AFT model achieved a higher log-likelihood than the Cox PH model empirically (Abadi et al., 2012; Sayehmiri et al., 2018).

Understanding the risk factors associated with readmission is the first step towards the goal of reducing avoidable readmission events. In recent years, machine learning algorithms have been widely used in readmission settings due to their superior predictive power in forecasting patients' risk scores. Hospitals usually prioritize resources to patients with higher risk scores. However, whether prioritizing high readmission risk patients reduces the readmission rate remains under debate. Recent researches have also shown that high readmission risk patients may be less sensitive to interventions than those at lower risk (Steventon & Billings, 2017). Due to the heterogeneity within individuals' responses to post-discharge care and other readmission interventions, more research on assessing readmission reduction strategies and personalized interventions is needed.

BIBLIOGRAPHY

Abadi, A., Amanpour, F., Bajdik, C., and Yavari, P. (2012). Breast cancer survival analysis: Applying the generalized gamma distribution under different conditions of the proportional hazards and accelerated failure time assumptions. *International journal of preventive medicine*, 3(9):644–651.

Bennett, K. J., Probst, J. C., Vyavaharkar, M., and Glover, S. H. (2012). Lower Rehospitalization Rates Among Rural Medicare Beneficiaries With Diabetes. *Journal of Rural Health*, 28(3):227–234.

Corrigan, J. M. and Martin, J. B. (1992). Identification of factors associated with hospital readmission and development of a predictive model. *Health services research*, 27(1):81–101.

Cox, D. R. (1975). Partial likelihood. *Biometrika*, 62(2):269–276.

Eby, E., Hardwick, C., Yu, M., Gelwicks, S., Deschamps, K., Xie, J., and George, T. (2015). Predictors of 30 day hospital readmission in patients with type 2 diabetes: A retrospective, case-control, database study. *Current Medical Research and Opinion*, 31(1):107–114.

Enomoto, L. M., Pradhan, D. P., Rosenthal, M., Hollenbeak, C. S., and Gabbay, R. A. (2016). Risk factors associated with 30-day readmission and length of stay in patients with type 2 diabetes. *Journal of diabetes and its complications*, 31(1):122–127.

Grambsch, P. M. and Therneau, T. M. (1994). Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*, 81(3):515–526.

Jiang, H. J., Andrews, R., Stryer, D., and Friedman, B. (2005). Racial/Ethnic Disparities in Potentially Preventable Readmissions: The Case of Diabetes. *American Journal of Public Health*, 95(9).

Klein, J. P. and Moeschberger, M. L. (2006). *Survival Analysis: Techniques for Censored and Truncated Data*. Statistics for Biology and Health. Springer, New York, NY, second edition.

Liang, Y., Chen, Z., Ward, R., and Elgendi, M. (2018). Hypertension Assessment via ECG and PPG Signals: An Evaluation Using MIMIC Database. *Diagnostics*, 8(3):65.

Naylor, M. D., Brooten, D., Campbell, R., Jacobsen, B. S., Mezey, M. D., Pauly, M. V., and Schwartz, J. S. (1999). Comprehensive discharge planning and home follow-up of hospitalized elders: A randomized clinical trial. *JAMA : the journal of the American Medical Association*, 281(7):613–620.

Ostling, S., Wyckoff, J., Ciarkowski, S. L., Pai, C.-W., Choe, H. M., Bahl, V., and Gianchandani, R. (2017). The relationship between diabetes mellitus and 30-day readmission rates. *Clinical diabetes and endocrinology*, 3(1):3–3.

Robbins, J. M. and Webb, D. A. (2006). Diagnosing Diabetes and Preventing Rehospitalizations: The Urban Diabetes Study. *Med Care*, 44(3):292–296.

Rogers, A. T., Bai, G., Lavin, R. A., and Anderson, G. F. (2017). Higher hospital spending on occupational therapy is associated with lower readmission rates. *Medical care research and review*, 74(6):668–686.

Rubin, D. J. (2015). Hospital Readmission of Patients with Diabetes. *Current Diabetes Reports*, 15(4).

Sandfort, V., Johnson, A. E., Kunz, L. M., Vargas, J. D., and Rosing, D. R. (2019). Prolonged Elevated Heart Rate and 90-Day Survival in Acutely Ill Patients: Data From the MIMIC-III Database. *Journal of Intensive Care Medicine*, 34(8):622–629.

Sayehmiri, K., Eshraghian, M. R., Mohammad, K., Alimoghaddam, K., Foroushani, A. R., Zeraati, H., Golestan, B., and Ghavamzadeh, A. (2008). Prognostic factors of survival

time after hematopoietic stem cell transplant in acute lymphoblastic leukemia patients: Cox proportional hazard versus accelerated failure time models. *Journal of experimental clinical cancer research*, 27(1):74–74.

Steventon, A. and Billings, J. (2017). Preventing hospital readmissions: the importance of considering 'impactibility,' not just predicted risk. *BMJ quality & safety*, 26(10):782–785.

Wei, N. J., Wexler, D. J., Nathan, D. M., and Grant, R. W. (2013). Intensification of diabetes medication and risk for 30-day readmission. *Diabetic Medicine*, 30(2).

Zhao, L., Zhao, L., Wang, Y. Y., Yang, F., Chen, Z., Yu, Q., Shi, H., Huang, S., Zhao, X., Xiu, L., Li, X., and Li, Y. (2020). Platelets as a prognostic marker for sepsis: A cohort study from the MIMIC-III database. *Medicine*, 99(45):e23151.

Appendix A

A SHORT NOTE ON R CODE

All statistical analyses in this thesis were conducted using R (version 4.0.0). The R code for data pre-processing, hypothesis testing, modeling, and graphing are available in my GitHub repository: `https://github.com/tq21/honors-thesis`

The SQL query for obtaining the data from the MIMIC-IV database is also provided in the repository, although access to the database is required and can be obtained directly from PhysioNet. A list of files and their descriptions is provided on the next page (Table A.1).

Table A.1: A list of files and descriptions

File name	Description
`clean.R`	code for data cleaning, including removing observations with unmatched discharge and readmission times, standardizing numerical covariates and assigning reference groups for categorical covariates
`eda.R`	code for hypothesis testing and summary statistics. `t.test`, `chisq.test`, and `wilcox.test` from 'stats' package (version 4.0.0); `survdiff` from 'survival' package (version 3.2-7).
`logistic_functions.R`	consists of functions that call the `glm` function from the 'stats' package; creates a `data.frame` object with the coefficient estimates and confidence intervals from the `glm` function output
`coxph_functions.R`	calls the `coxph` function from the 'survival' package; the function `cox.zph` from the 'survival' package was used to test the proportional hazards assumption of the Cox PH model
`cox_aft.R`	constructs the Cox PH model; calls the `flexsurvreg` function from the 'flexsurv' package (version 1.1.1) to build the AFT model
`figures.R`	code for creating the tables and figures in this thesis
`aft_multi.R`	variable selection, fitting the multivariate AFT model

Appendix B

ACCELERATED FAILURE TIME MODEL DIAGNOSTICS

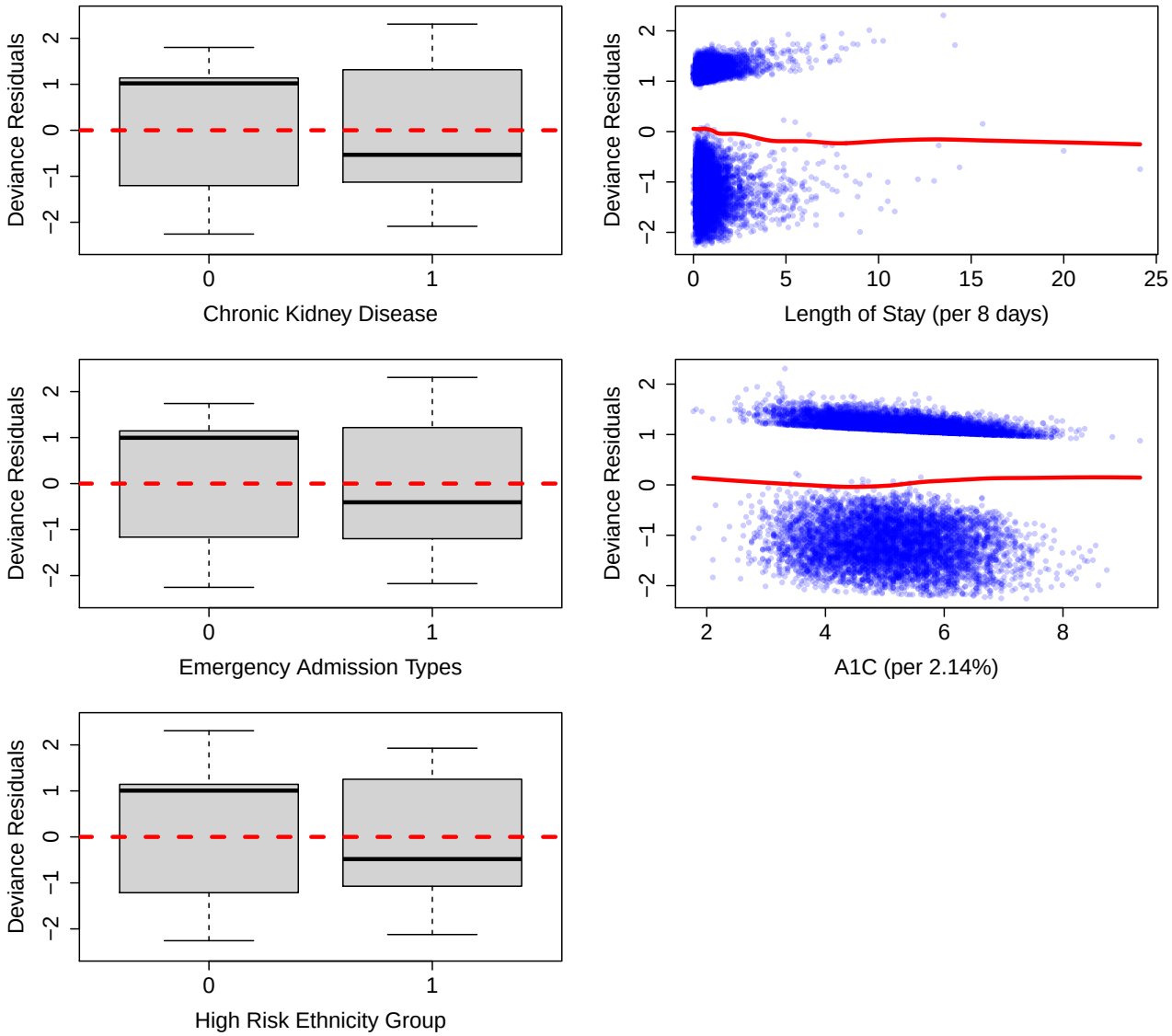



Figure B.1: Deviance residual plots